Financial News in Indonesian Language Multi-Class Sentiment Classification Using R Keras/TensorFlow

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Abstract

Financial markets react strongly to news sentiment. Traders, analysts, and automated trading systems increasingly use NLP methods to quantify news sentiment and integrate it into trading or risk-monitoring workflows. For this project, I developed a complete Natural Language Processing (NLP) pipeline to classify the sentiment of financial news articles from CNBC Indonesia into three categories: positive, neutral, and negative. I focused on financial market news in Indonesian language, my native language, as such analysis is not yet available widely in the Indonesia market. To make it beyond our assignment, I moved from binary spam classification to multi-class sentiment analysis (positive, neutral and negative). And furthermore, I did 2 iterations, one with all stock news and second with only the top 45 Indonesian big cap stocks, which is widely known as LQ45. It is common for financial market analysts to analyse for market index and for only a selected group of stocks. Creating a subset of big cap stocks is aligned with market practice and useful for traders, analysts and trading systems.
I used TF–IDF + logistic regression from the classical model for the baseline model and Embedding + Bidirectional LSTM (BiLSTM) from the sequence model for the neural model, and compared their performance on both whole market stock news and subset of big cap stocks news.

For the dataset, I used the CNBC Indonesia Stock News Sentiment Dataset. This dataset contains 9,819 Indonesian stock market news headlines published on CNBC Indonesia. Each headline has been manually labeled with one of three sentiment classes: positive, neutral and negative.

The final results show that, for all market news, the baseline model achieves **65.4%** accuracy, while the neural model improves performance to **80.1%**, demonstrating the benefit for financial sentiment in Indonesian language. But, for the subset of big cap stock news, the baseline model achieves **72.4%** while the neural model achieves **71.0%**. When working with smaller datasets, simple TF–IDF + logistic regression might perform better due to risk of overfitting and other possible downsides from deeper neural models.

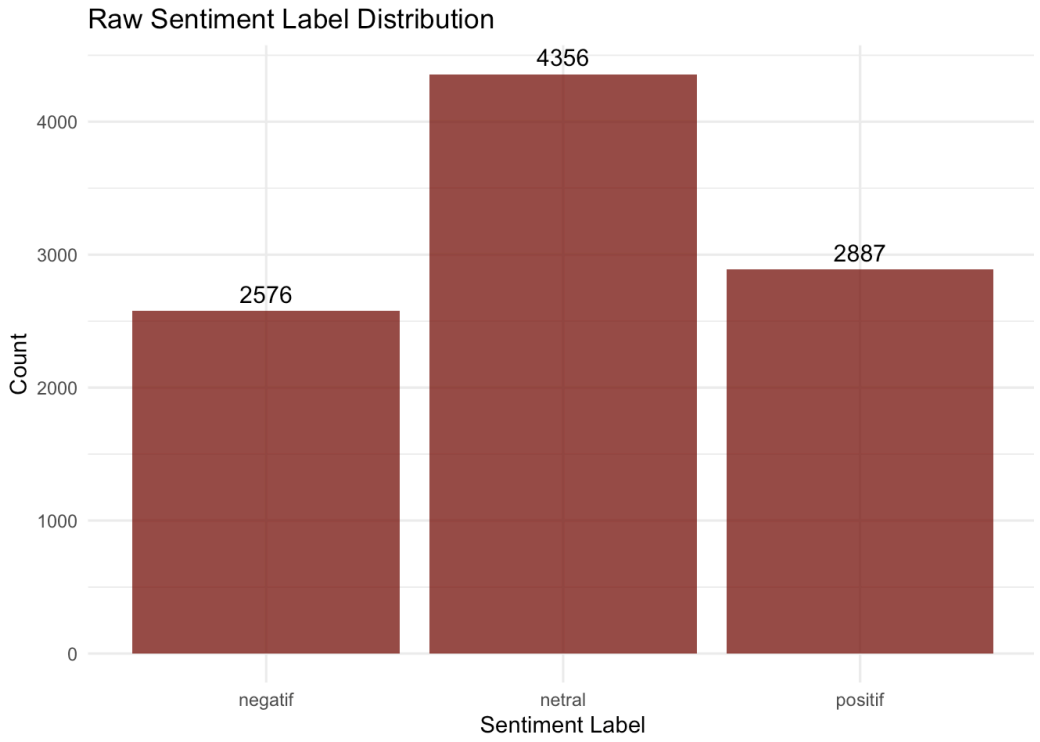
For future improvement, we could work with bigger datasets consisting of more news not only from news outlets but also from social media contents on the financial market. We could also reduce neural model capacity when dealing with subsets of stocks news. Furthermore, we could consider pre-trained Indonesian embeddings or IndoBERT as pre-trained language models often outperforms both models on limited data.

Problem Statement

Financial markets react strongly to news sentiment. Traders, analysts, and automated trading systems increasingly use NLP methods to quantify news sentiment and integrate it into trading or risk-monitoring workflows. In Indonesia, financial market news sentiment analysis is not yet widely available due to the lag of market and technology sophistication in the market. As such, I would like to do a model for doing sentiment analysis for Indonesian stock market news.

Dataset Description

This dataset contains 9,819 Indonesian stock market news headlines published on CNBC Indonesia under the "Market" category from January 1, 2024 to March 31, 2025. Each headline has been manually labeled with one of three sentiment classes: positive, neutral, or negative. The dataset is obtained from Kaggle (https://www.kaggle.com/datasets/triagungj/cnbc-indonesia-stock-news-sentiment-dataset).



The datasets contained 4,356 neutral, 2,887 negative and 2,576 positive stock news in Indonesian language.

Methodology

For this project, I used the classical model TF–IDF + logistic regression for the baseline model and sequence model Embedding + Bidirectional LSTM (BiLSTM) for the neural model. I used the models on all stock news and then on only a subset of big cap stocks news. For big cap selection, I used the big cap Indonesian stock index, namely LQ45, which consists of the top 45 Indonesian stocks by market capitalization. After that, I compared their performances by analyzing the accuracy and confusion matrices.

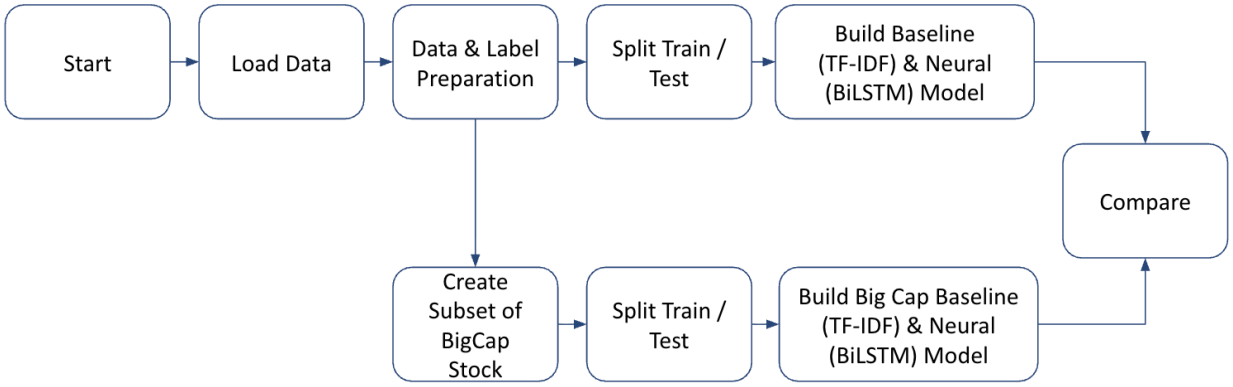
Installation & Configuration

I started by setting the necessary library below.

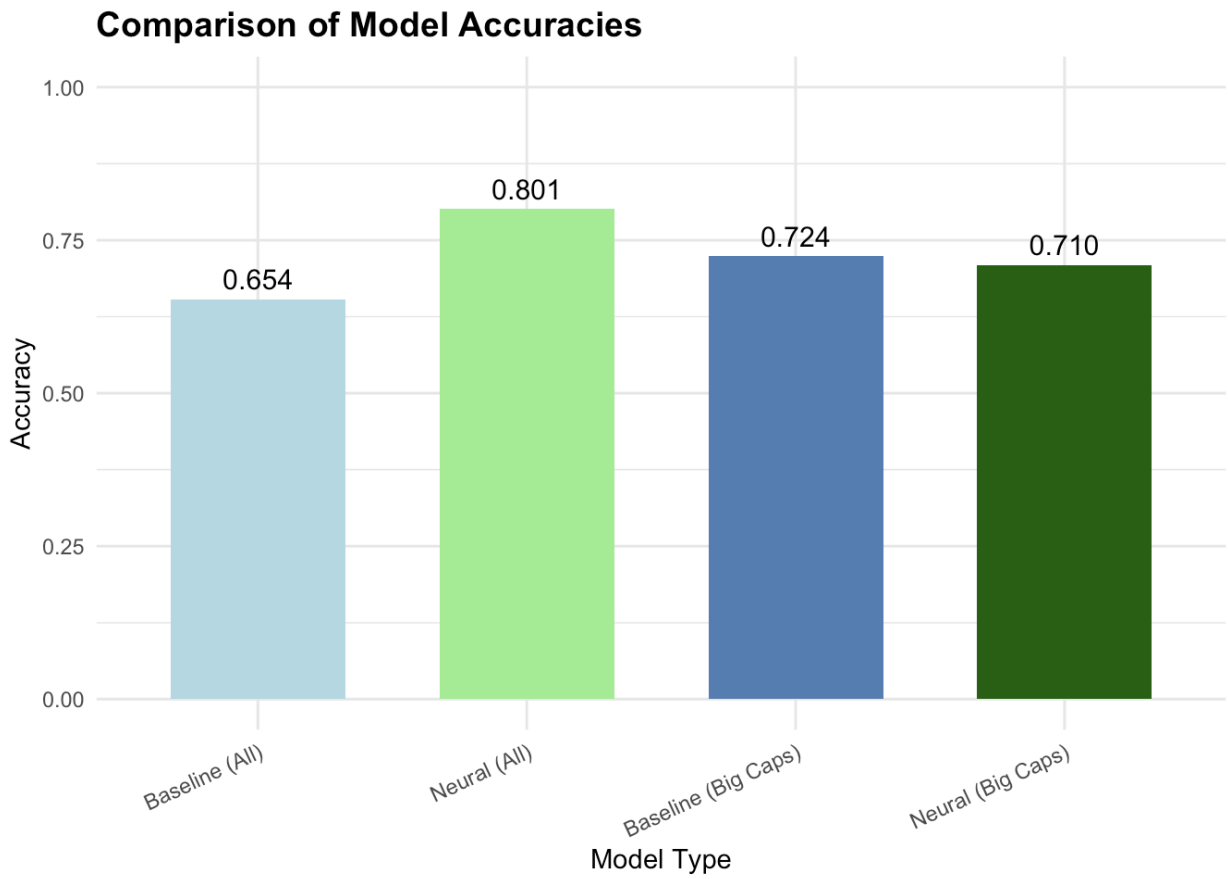
```
library(dplyr)
library(ggplot2)
library(caret)
library(stringr)
library(tm)
library(glmnet)
library(keras)
library(tensorflow)
# install_keras() #install if needed

set.seed(123)
```

I continued with loading and preparing the datasets, continued by training and testing the baseline and neural model on all stock dataset and on big cap (LQ45) stock datasets as per the diagram below.



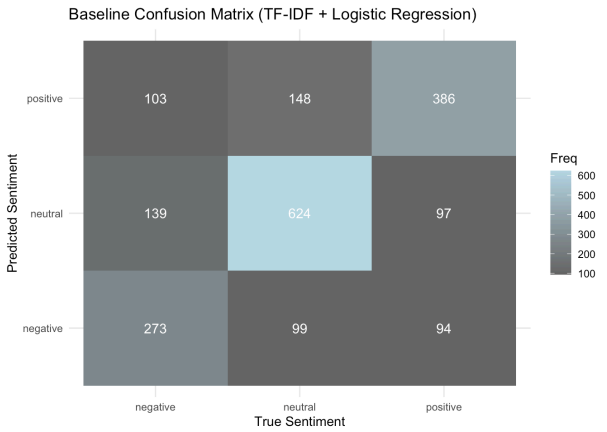
Results

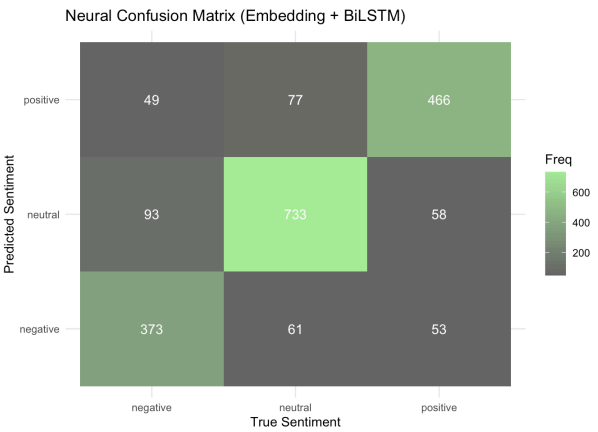


For all stock news dataset, the neural model achieved accuracy of **80.1%** and performed better than the baseline model with **15%** improvement. The accuracy is considered respectable and can be useful to be used in financial market analysis for traders, analysts or automated trading systems. This improvement is meaningful and aligns with what we typically observe when moving from a linear bag-of-words model to a sequence-aware neural network.The Embedding + BiLSTM is able to capture context, ordering, and indirect sentiment expressions that TF-IDF + logistic regression cannot, particularly in financial text where polarity is often implied rather than explicitly stated.

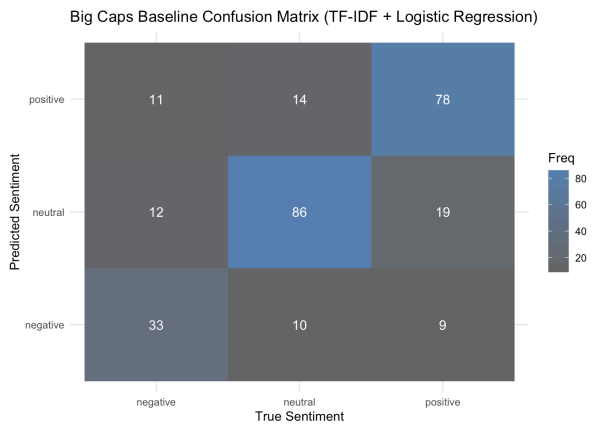
Nevertheless, when analyzing certain market indices that consist only of a certain set of stocks, using a simpler model can be more useful, as indicated by the accuracy numbers above. The big cap stocks subset is narrower in scope, more homogeneous in vocabulary, and more structured around recurring financial reporting language (e.g., earnings calls, price movements, regulatory updates). In such cases, TF-IDF + logistic regression tends to perform relatively well because sentiment words and surface forms are more consistent.

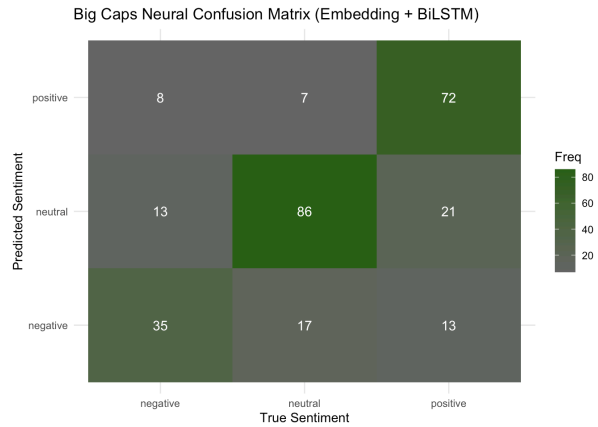
All Stock Baseline vs Neural Confusion Matrices:





Big Cap Stock Baseline vs Neural Confusion Matrices:





From the confusion matrices, we can learn that neural models outperform baseline models when the dataset is broad and heterogeneous. When dealing with LQ45 big cap index, the reporting is more predictable, and thus producing better baseline performance and a narrower performance gap between baseline and more complex neural approaches.
From the number of sentiments, neutral sentiment is easiest to classify. It is due to its frequency and predictable linguistic patterns in financial reporting. Negative sentiment is consistently the most difficult, which can be attributed to the subtlety of negative tone in Indonesian context, where negative information is often presented with cautious or softened language. Lastly, we can see the baseline's tendency to misclassify negative items as positive. Sequence models can reduce extreme polarity confusion as we can see the improvement in false positives from baseline to neural model.

Conclusion & Future Improvement

Embedding + BiLSTM worked for stock news even in non-English language. It performed better than TF-IDF + logistic regression in financial text where polarity is often implied rather than explicitly stated. When dealing with certain subsets of stocks such as certain indices, one must expand the datasets or check the accuracy as there is a risk of overfitting when using BiLSTM. In such cases of more focused datasets, a simpler model tends to perform relatively well because sentiment words and surface forms are more consistent in certain groups of stocks.

For future improvement, we could work with bigger datasets consisting of more news not only from news outlets but also from social media contents on the financial market. We could also reduce neural model capacity when dealing with subsets of stocks news. Furthermore, we could consider pre-trained Indonesian embeddings or IndoBERT as pre-trained language models often outperforms on limited data.